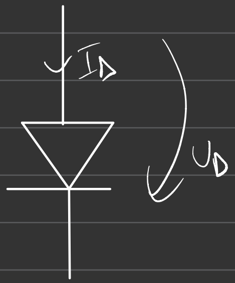
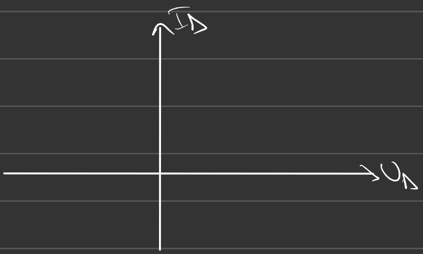
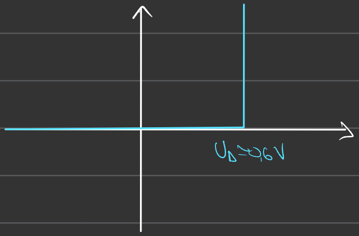
Curs 4



Dioda ideală



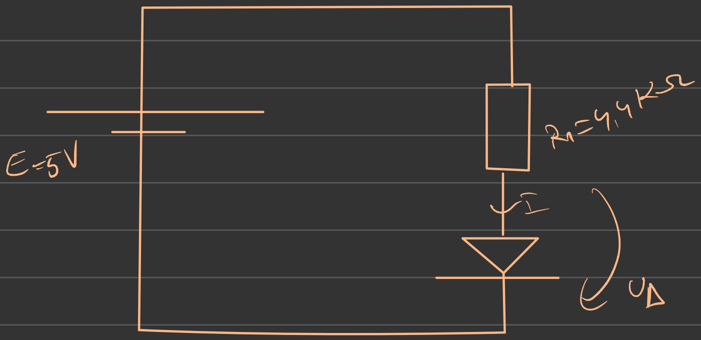
Tensiune constantă



$$\begin{cases} U_D < 0, & I_D = 0 \\ I_D > 0, & U_D = 0 \end{cases}$$

$U_D < 0{,}6V$ $I_D = 0$ Blocată

$I_D > 0$ $U_D = 0{,}6V$ Conducție



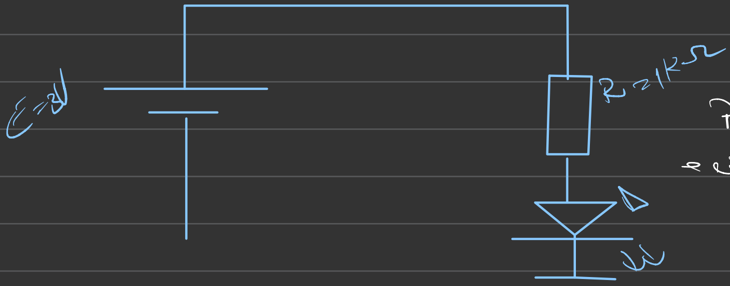
$U_D = 0{,}6V$
(conducție)

$I = ?$

Presupunem că dioda este în conducție

KVL KII $E = IR_1 + U_D$

$$I = \frac{E - U_D}{R_1} = 1mA > 0 \quad \text{(PP adev)}$$



Presupunem că Dioda e în conducție => $U_D = 3{,}3V$

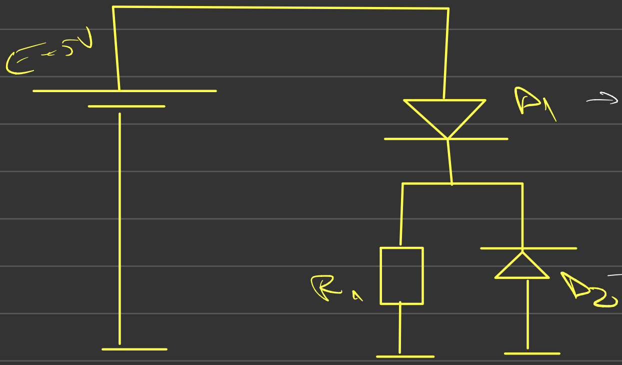
$$E = IR + U_D$$

$$I = \frac{E - U_D}{R} = -0{,}3 \text{ mA}$$

(PP greșit)

=> Dioda e blocată => $I = 0$ A

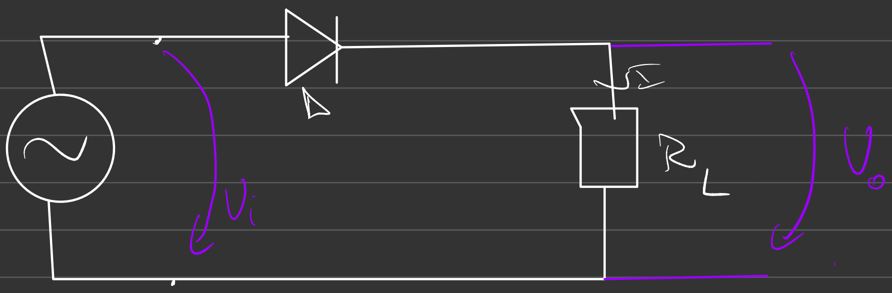
$U_D = 0{,}6$ V

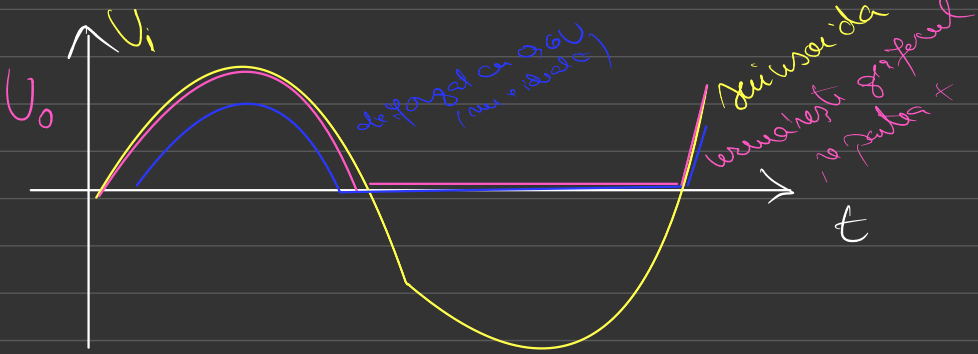


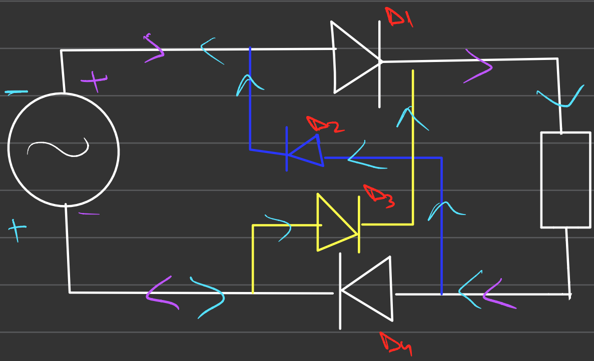
D_1 -> conducție

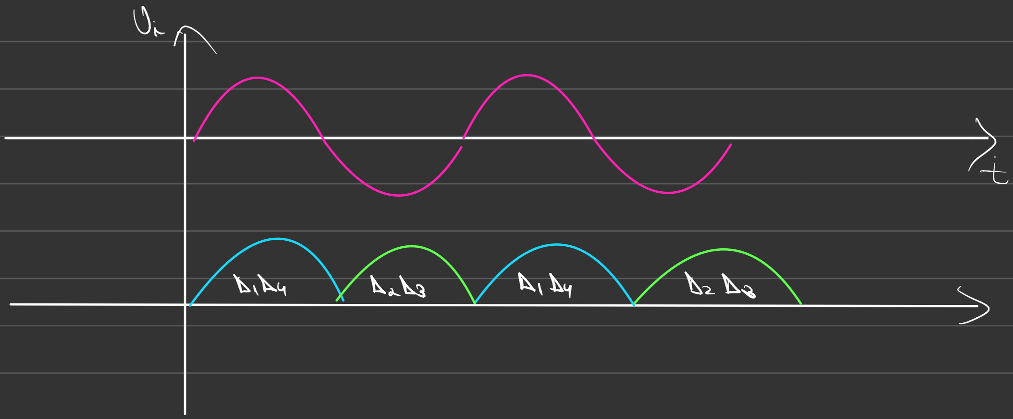
D_2 -> blocată

REDRESOR MONOALTERNANT

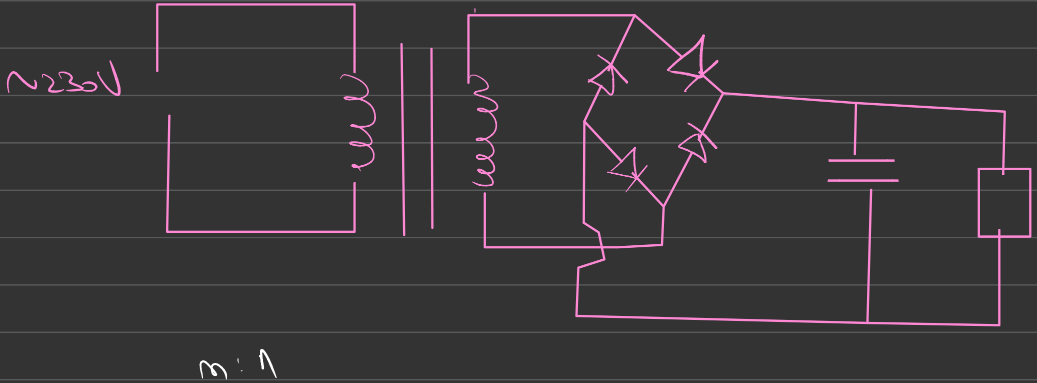


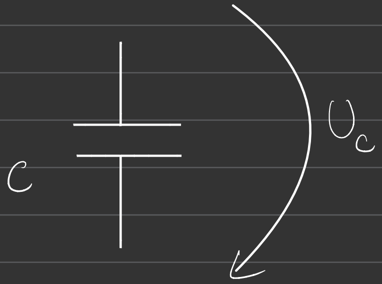




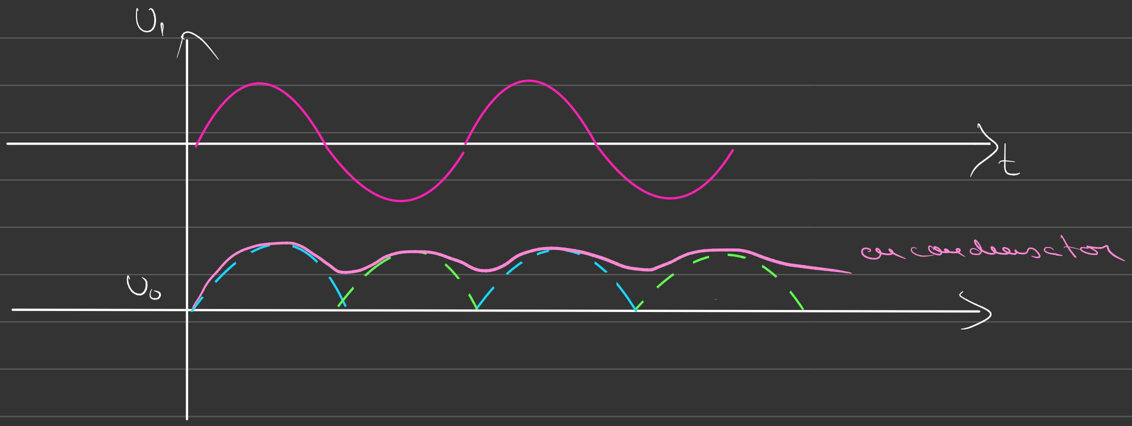


Redresor bialternant

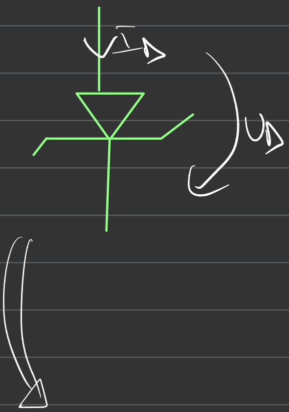


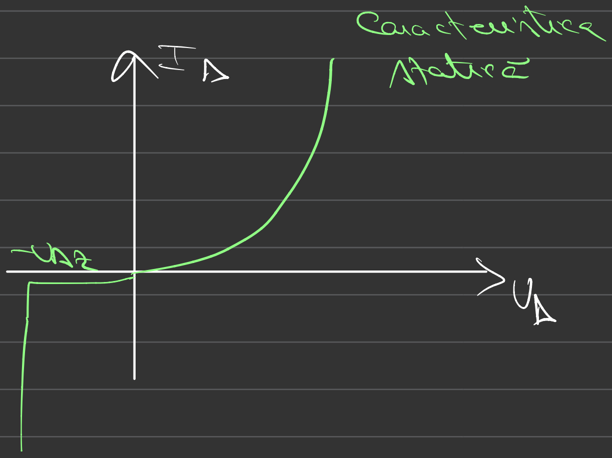


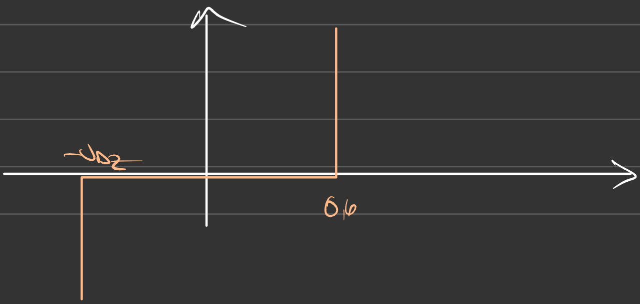
$$\boxed{I_c = C\frac{dU_c}{dt}}$$



DIODA ZENER





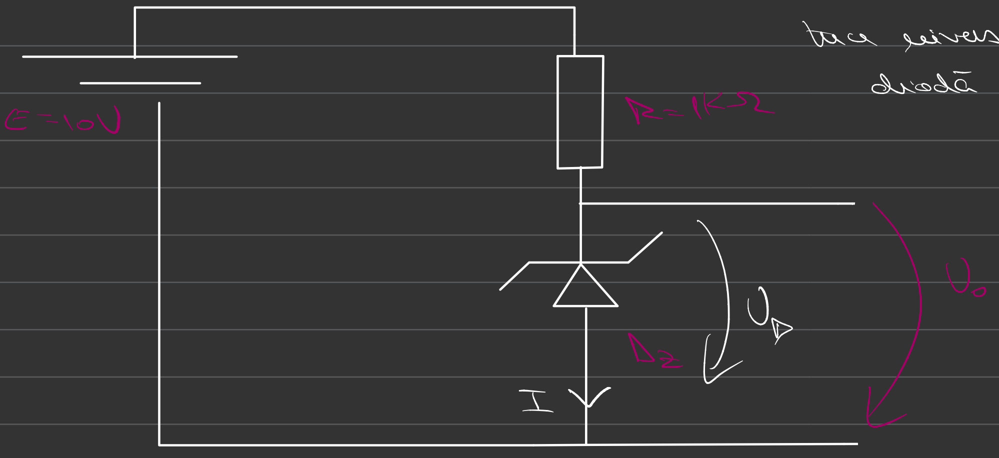


$I_D > 0$, $U_D = 0{,}6V$ conducție

$-U_{DZ} < U_D < 0{,}6$, $I_D = 0$ blocată

$I_D < 0$, $U_D = -U_{DZ}$ conducție inversă

Aici curentul trece invers prin diodă


E=10V
R=1KΩ
U_D
U_0
D_Z
I

$U_D = 0{,}6$ V (cand directă)

$U_{DZ} = 5$ V

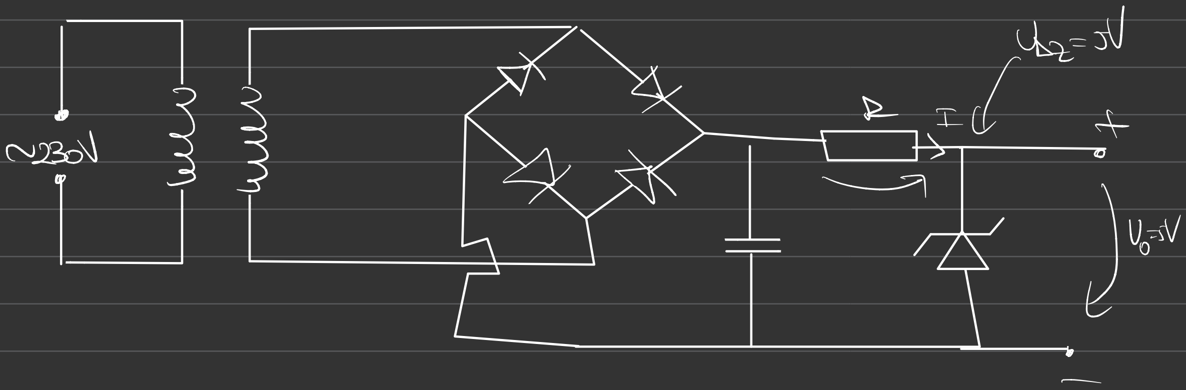
$U_0 = ?$

Pp că D_Z = cond inversă

$I_D > 0$, $U_D = U_{DZ}$

$E = RI + U_{DZ} \Rightarrow I = \frac{E - U_{DZ}}{R} = 5mA$

$U_0 = U_D = 5V$


~230V
R
I
$U_{DZ}=5V$
+
$U_0=5V$
−